

JIANGSU BAOLING CHEMICAL CO., LTD.

NO.9 Tongwang Road, Nantong, Jiangsu, China

Tel: 86-513-83513256,85056008; Fax: 86-513-83518952

MATERIAL SAFETY DATA SHEET

1. Product And Company Identification:

Product: Metalaxyl 25% WP

Company Identification: Jiangsu Baoling Chemical Co., Ltd.

No. 9 Tongwang Road, Nantong, Jiangsu, China

Use: Fungicide

2. Composition/Information On Ingredients:

Ingredients	Conc. (%)
Metalaxyl Tech	25
Other ingredient	75

3. Hazardous Identifications:

Component	Sympol	R phrases
Metalaxyl	N	R22/52

4. First Aid Measures:

Have the product container, label or Material Safety Data Sheet with you when you calling a Poison control center or doctor, or going for treatment.

Ingestion: If swallowed, call a Poison control center or a Doctor immediately for treatment advice. Have the person sip a glass of water if able to swallow. Do not induce vomiting unless to do so by a Poison Control center or Doctor. Do not give anything by mouth to an unconscious person.

Eye: If in eyes, Hold eye open and rinse slowly and gently with water for 15 –20 minutes. Remove contact lenses, if present, after 5 minutes, then continue rinsing eye. Call a poison control center or doctor for treatment advice.

Skin: Wash affected area with plenty of soap and water for 15 – 20 minutes. Remove contaminated clothing. Launder contaminated clothing separately. Get medical attention.

Inhalation: Move person to fresh air. If person is not breathing give artificial respiration, preferably mouth to mouth if possible. Call a Poison control center or doctor for further treatment advice.

Note to Physician: There is no specific antidote if this product is ingested. Treat symptomatically.

5. Fire Fighting Measures:

Unusual Fire, Explosion and Reactivity Hazards: During a fire, irritating and possibly toxic gases may be generated by thermal decomposition or combustion.

In case of Fire: Use dry, chemical, foam or CO₂ extinguishing media. Wear full protective clothing and self-contained breathing apparatus. Evacuate nonessential personnel from the area to prevent human exposure to fire, smoke, fumes or products of combustion.

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Prevent use of contaminated buildings, area and equipment until decontaminated. Water runoff can cause environmental damage. If water is used to fight fire, dike and collect runoff.

6. Accidental Release Measures:

Control the spill at its source. Contain the spill to prevent from spreading or contaminating soil or from entering sewage and drainage systems or any body of water.

Clean up spills immediately, observing precautions in protective equipment section. Cover entire spill with absorbing material and place into compatible disposal container. Scrub area with hard water detergent (e.g. commercial products such as Tide, Joy, Spic and Span). Pick up wash liquid with additional absorbent and place into compatible disposal container. Once all material is cleaned up and placed in a disposal container, seal container and arrange for disposition.

7. Handling and Storage:

Handling: Keep Out Of Reach Of Children and animals. Prevent eating, drinking, tobacco use, and cosmetic application in areas where there is a potential for exposure to the material. After work, rinse gloves and remove protective equipment. Wash hands thoroughly with soap and water after handling, and before eating, tobacco use, drinking, or using the toilet. Wash contaminated clothing before re-use and separate from household laundry. Keep containers closed when not in use. Keep product, wash or rinse water, and contaminated materials out of water, away from crops, and away from access by people, animals and birds.

Storage: Store in original container only in a well-ventilated, cool, dry, secure area. Protect from heat, sparks and flame. Do not expose sealed containers to temperatures above 40 °C and prevent product from freezing. Keep separate from other products to prevent cross contamination. Rotate stock. Clean up spilled material immediately.

8. Exposure Control/Personal Protection:

Applicable control measures, including engineering controls: Please refer to the product label for personal protective clothing/equipment. Personal protective equipment for each exposure route:

Ingestion: Do not eat, drink, smoke or apply cosmetics in areas where there is a potential for exposure to the material. Always wash thoroughly after handling.

Eye Contact: To avoid eye contact, wear safety glasses with side shields or chemical goggles.

Skin Contact: To avoid skin contact, wear rubber gloves, rubber boots, long-sleeved shirt, long pants and a head covering.

Inhalation: To avoid breathing dust, wear pesticide respirator with particulate profilers. Select N or R or P type as appropriate for the oil characteristics of any other air contaminants present. Filter efficiency ranges from 95-99.7% as appropriate for the size distribution of dust present.

Applicators: Please refer to product label for personal protective clothing and equipment.

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9. Physical and Chemical Properties:

<i>Appearance:</i>	white powder
<i>Melting Point:</i>	63.5 – 72.3°C (Tech)
<i>Density:</i>	1.20 @20°C
<i>Water solubility:</i>	8.4 g/l (22°C)
<i>Other solubility:</i>	Soluble in ethanol 400, acetone 450, toluene 340, <i>n</i> -hexane 11, <i>n</i> -octanol 60 (all in 25°C)
<i>pH value:</i>	5-8

10. Stability and Reactivity

Chemical stability: Stable under normal conditions of use.

Hazardous polymerisation: Will not occur

Conditions to avoid: Extreme heat and fire

Incompatible materials: None known

Hazardous decomposition products: Thermal decomposition may produce toxic by-products of carbon and nitrogen.

11. Toxicological Information

Contact with the skin: Irritate to skin

Contact with the eyes: Eyes become red

Inhalation: Irritate to respiratory tract

Ingestion: nausea, vomiting and breathing difficulty.

Sharp toxicity: not applicable

Effects for chronic exhibition

Chronic toxicity: A 90-day study of rats exposed to 0.1 to 2.5 mg/kg/day in diet, showed some cellular enlargement in the liver at the highest dose. In a similar study with dogs fed diets of approximately 0.04 to 0.8 mg/kg/day for 6 months, the dogs were adversely affected by the highest dose. Manifestations included increase blood alkaline phosphatase and increased liver-to-brain weight ratio.

Reproductive effects: A three-generation rat study where animals were fed up to 2.5 mg/kg/day showed no compound related maternal toxicity or reproductive effects. These data suggest that metalaxyl is unlikely to cause reproductive effects.

Teratogenic effects: Rats given a dosage of 120 mg/kg/day by stomach tube on days 6 to 15 of gestation exhibited no embryotoxicity or teratogenicity, nor did rabbits given a dosage of 20 mg/kg/day by the same route on day 6 to 8. These data suggest that metalaxyl is not teratogenic.

Mutagenic effects: studies including a dominant lethal assay in male mice indicate that metalaxyl has no mutagenic potential.

Carcinogenic effects: Available studies of the carcinogenicity of metalaxyl are inconclusive

Organ toxicity: The liver is the primary target organ for metalaxyl in animal systems

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Sensitization: non-sensitive

12. Ecological Information:

Effect on birds: moderate toxicity to birds, acute oral LD₅₀ for Mallard ducks is 1466 a.i.mg/kg.

Effect on fish: moderate toxicity to fish, acute 96 hour LC₅₀ for Rainbow trout is 0.96 a.i.mg/L.

Effect on aquatic invertebrates: moderate toxicity to aquatic invertebrates, acute 48 hour

EC₅₀ for *Daphnia magna* is 3.47 a.i.mg/L.

Effect on algae: moderate toxicity to algae, acute 72 hour EC₅₀ for *Scenedemus subspicatus* is 0.42 a.i.mg/L.

Effect on honeybees: low toxicity to honeybees, contact acute 48 hour LD₅₀ is 200 a.i.µg/bee; oral acute 48 hour LD₅₀ is 269 a.i.µg/bee.

Effect on earthworms: low toxicity to earthworms, acute 14 day LC₅₀ for *Eisenia foetida* is >1000 a.i.mg/kg.

13. Disposal Considerations

Triple or (preferably) pressure rinse containers before disposal. Add rinsings to the spray tank. Do not dispose of undiluted chemicals on-site. If recycling replace cap and return clean containers to recycler or designated collection point. If not recycling, break, crush or puncture and bury empty containers in a local authority landfill. If no landfill is available bury the containers below 500 mm in a disposal pit specifically marked and set up for this purpose clear of waterways, desirable vegetation and tree roots. Empty containers and product should not be burnt.

14. Transport Information

Do not transport with food and feedstuff.

Proper shipping name: ENVIRONMENTAL HAZARDOUS SUBSTANCE,
SOLID, N.O.S. (Metalaxyl 25%)

UN No. 3077

Class: 9

15. Regulatory Information:

Symbol: N (Dangerous for environment)

R phrases:

R22: Harmful if swallowed;

R52: Harmful to aquatic organisms;

S Phrases:

S1/2: Keep locked up and out of the reach of children;

S3/7: Keep container tightly closed in a cool place;

S36/37/39: Wear suitable protective clothing, gloves and eye/face protection;

16. Other Information:

N/A